

Hackathon use case: Talk to data – Seamless self-service intelligence

Build solutions using free-tier AI of your choice that can work with datasets and make the process self-service to aid re-use.

Problem statement

Many people struggle to get quick, accurate, and trustworthy answers from data.

They face challenges such as:

- Too many steps or complicated tools.
- Unclear terminology.
- Pressure to respond fast.
- Lack of confidence in the data.

Your task is to remove friction and help everyday users “talk” to data effortlessly, receiving answers they can trust.

Description

Create a solution where users can ask natural language questions about any dataset and instantly receive:

- Clear explanations.
- Verified insights.
- Transparent data sources.
- Minimal steps.
- No exposure of private or sensitive data.

Focus on the three pillars:

1

Clarity:

Answers simple enough for nonexperts.

2

Trust:

Clear definitions, consistent metrics, and source transparency.

3

Speed:

Near instant responses without complex workflows.

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Use cases



1. Understand what changed

Users ask:

"Why did revenue drop last month?"
"What caused customer complaints to rise?"

The AI system should:

- Identify the drivers behind increases or decreases.
- Highlight the most influential categories (e.g., region, product, channel).
- Provide clear, concise explanations in everyday language.
- Reference data sources used for the insight.

Output example:

Revenue decreased by 11% in Feb. The biggest contributor was a 22% drop in the South region due to reduced ad spend.

2. Compare (time, region, product, segment)

Queries may include:

"This week vs last week"
"Region A vs Region B"
"Product X vs Product Y performance"

The AI should:

- Interpret the comparison intent.
- Apply consistent metric definitions.
- Generate an easy-to-understand comparison (visual + text).
- Identify statistically relevant differences.
- Handle ambiguous phrases like "this month" or "last cycle".

Output example:

Product A grew by 8% WoW, outperforming Product B (+2%). Primary reason: higher return customer rate.

3. Breakdown (decomposition)

Users ask:

"What makes up total sales?"
"Show the breakdown of costs by department."

The system should:

- Decompose a number into its components (by region/category/product/channel).
- Surface patterns (e.g., concentration, outliers).
- Highlight the biggest contributors.
- Provide both table and narrative explanation.

Output example:

North region accounts for 40% of total sales, with Retail contributing most of that share.

4. Summarize (daily/weekly/monthly insights)

A simple prompt such as:

"Give me a weekly summary for customer metrics."

The AI should:

- Scan datasets for trends, anomalies, and important shifts.
- Produce a concise update.
- Avoid noise – focus on what truly matters.
- Provide source references.
- Make summaries easy for leadership and nontechnical users.

Output example:

This week: Signups grew by 5%, churn remained stable, and average handle time improved by 12 seconds.

Learning outcomes

Participants will learn:

1. How to turn everyday questions into trustworthy answers?

- Translating vague or informal queries into structured analytical tasks.
- Handling ambiguous or incomplete questions safely.
- Providing grounded, data backed responses.

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2. Why shared definitions matter for consistent results?

- Building a semantic layer/metric dictionary.
- Ensuring “revenue”, “orders”, “active users” always mean the same thing.
- Preventing misalignment across teams.

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3. How to explain results clearly to non-experts?

- Removing jargon.
- Using simple language.
- Surface first summaries with deeper layers available.
- Showing data sources and reasoning transparently.



References

- Chat with your database the right way: collaborating agents.
- Snowflake Intelligence: Talk to Your Data, Unlock Real Business Insights.
- Build an AI-powered text-to-SQL chatbot using Amazon Bedrock, Amazon MemoryDB, and Amazon RDS | AWS…

